

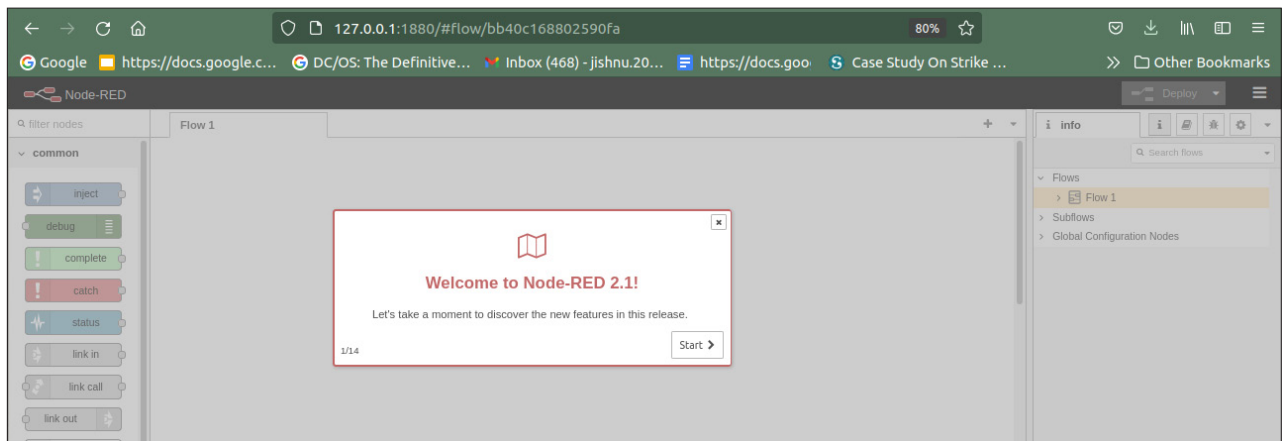


Figure 4: Node-RED in the browser

```
jishnusaurav@jishnusaurav-IdeaPad-L340-15IRH-Gaming:~$ node-red
23 Nov 23:08:31 - [info]

Welcome to Node-RED
=====

23 Nov 23:08:31 - [info] Node-RED version: v2.1.3
23 Nov 23:08:31 - [info] Node.js version: v14.4.0
23 Nov 23:08:31 - [info] Linux 5.4.0-90-generic x64 LE
23 Nov 23:08:31 - [info] Loading palette nodes
23 Nov 23:08:32 - [info] Settings file : /home/jishnusaurav/.node-red/settings.js
23 Nov 23:08:32 - [info] Context store : 'default' [module=memory]
23 Nov 23:08:32 - [info] User directory : /home/jishnusaurav/.node-red
23 Nov 23:08:32 - [warn] Projects disabled : editorTheme.projects.enabled=false
23 Nov 23:08:32 - [info] Flows file : /home/jishnusaurav/.node-red/flows.json
23 Nov 23:08:32 - [info] Creating new flow file
23 Nov 23:08:32 - [warn]

-----
Your flow credentials file is encrypted using a system-generated key.

If the system-generated key is lost for any reason, your credentials
file will not be recoverable, you will have to delete it and re-enter
your credentials.

You should set your own key using the 'credentialSecret' option in
your settings file. Node-RED will then re-encrypt your credentials
file using your chosen key the next time you deploy a change.
-----

23 Nov 23:08:32 - [info] Server now running at http://127.0.0.1:1880/
23 Nov 23:08:32 - [info] Starting flows
23 Nov 23:08:32 - [info] Started flows
```

Figure 5: Adding the script

Now, we need to install NPM (node package manager).

```
sudo apt-get install npm
```

To check the installation, use the following command:

```
npm -v
```

The next step is to install a helping utility called *node-red-admin*, using the following function, as shown in Figure 3.

```
sudo npm install -g --unsafe-perm node-red node-red-admin
```

We now need to allow one of our firewall ports. Since 1880 is a comparatively less used port, let us proceed using the following command:

```
sudo ufw allow 1880
```

Finally, we can start Node-RED by just entering it on the terminal, as shown in Figure 4. There will now be a link out, which probably is *http://127.0.0.1:1880/*. We can now use Node-RED in the browser, as shown in Figure 4.

A small script needs to be added to start Node-RED automatically on startup. Use the following command for that:

```
sudo nano /etc/systemd/system/node-red.service
```

Once the editor of the *.service* file opens up, enter the following script, as shown in Figure 5.

```
[Unit]
Description=Node-RED
After=syslog.target network.target
```

```
[Service]
ExecStart=/usr/local/bin/node-red-pi
--max-old-space-size=128 -v
Restart=on-failure
KillSignal=SIGINT
```

```
# log output to syslog as 'node-red'
SyslogIdentifier=node-red
StandardOutput=syslog
```

```
# non-root user to run as
WorkingDirectory=/home/sammy/
User=sammy
Group=sammy
```

```
[Install]
WantedBy=multi-user.target
```

To check if our Node-RED is still working properly, we can use the following command:

```
sudo systemctl start node-red
```

If there are no errors, it means that everything has been set up perfectly.